



CIRCULAIRE STROMEN **NEXT LEVEL LOGISTICS**

Towards sustainable logistics





CS-NLL PRESENTATION

MONDAY NOVEMBER 18TH 2024

Bas Holland
Business development LCB |
Projectcoördinator CS-NLL

CIRCULAIRE STROMEN
NEXT LEVEL LOGISTICS

CONTENT PRESENTATION

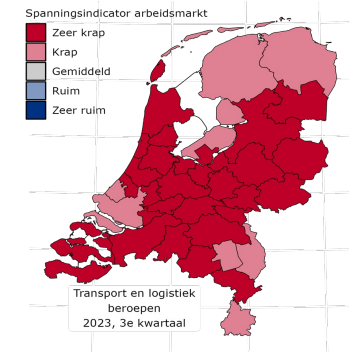
1. General developments with impact (on logistics)
2. Trends in logistics
3. Logistics Community Brabant
4. Living Innovation Lab projects
5. Running example project "Circulaire Stromen (circular flows) – Next Level Logistics"
6. Example of Data Science & AI student assignments

GENERAL DEVELOPMENTS WITH IMPACT (ON LOGISTICS)

1. Climate change and emerging legislation
2. Global disruptions
3. Aging of working capital
4. Urbanization – household composition

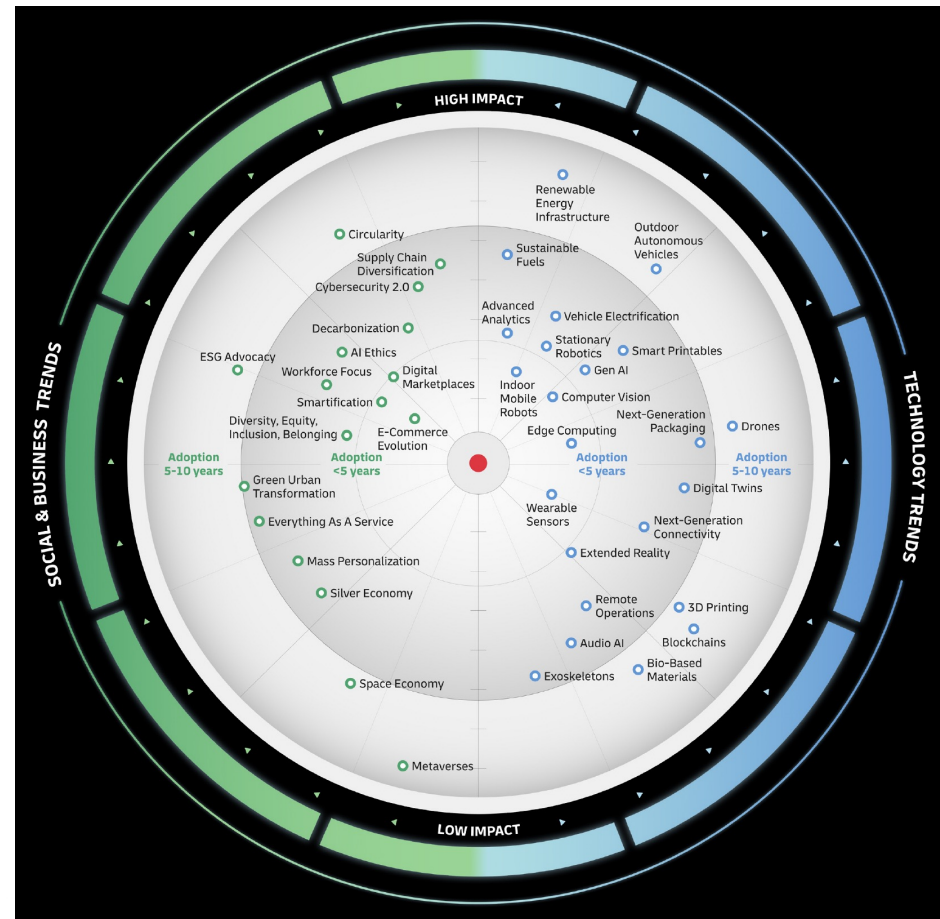
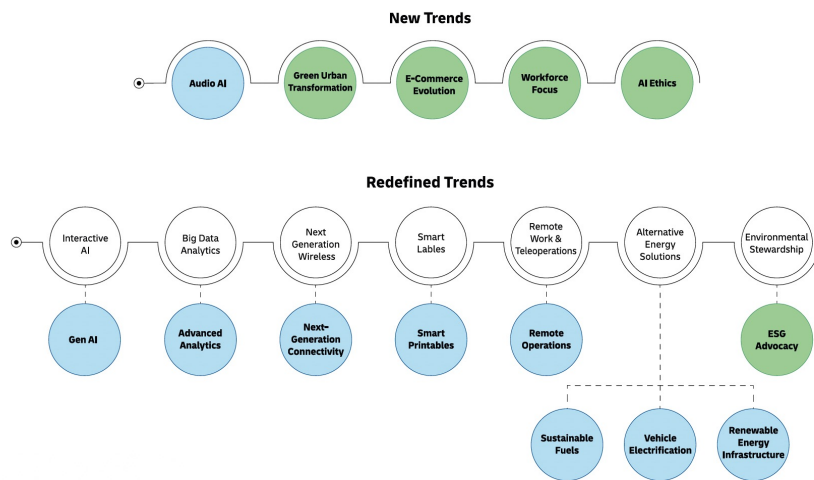
And.... the enabling role of logistics for society

- circular economy
- local energy hubs (roofs warehouses)
- living independently at home longer (product availability)
- world food program



TRENDS IN LOGISTICS

The DHL Logistics Trend Radar 7.0



CIRCUAIRE STROMEN
NEXT LEVEL LOGISTICS

LOGISTICS COMMUNITY BRABANT - LCB



**INNOVATION
TOGETHER,
TODAY AND
TOMORROW**

Logistics Community Brabant

Network organization of knowledge institutions (BUas, TIU, TU/e and NLDA) that aims to bring knowledge, innovation and talent to the logistics field more quickly and easily.

- **160.000** residents in Brabant working in the logistics sector
- **1.000** students (Fresh Brains) who are following a logistics education at higher professional or university level
- **500** experts in the LCB network with specific (supply chain) expertise.

LCB MEMBERS

PROJECTS

CONTRACT ACTIVITIES

Community's

- Transport & warehousing
- Industry
- Livable city
- Event logistics
- Care

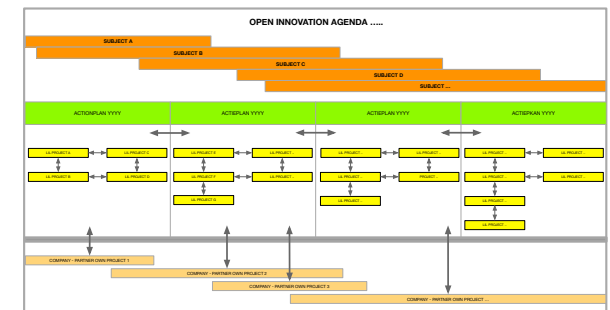
DIGITALIZATION

SUSTAINABILITY

- Logistics is all about the moving of everything and everyone.
- Transformation from diesel to data.
- At LCB (trustworthy) sharing is key.

LIVING INNOVATION LAB PROJECTS

1. Executed in a real-live setting in a partnership between knowledge institutions and companies.
2. Open innovation agenda (outside-in) + (periodic – annual) actionplan.
3. Focus on proof of concepts (POC's) → TR6 / TR7.
4. Focus on changemanagement (employee acceptance).
5. Mix of employees, teachers – professors, experts and students.
6. Relay principe (student waves) → stacking & connecting.
7. Subsidized projects or LCB member community's.
8. Sharing is key →
 - between living innovation labs projects and community's;
 - curriculum enrichment;
 - sector valorization.



TRL 9	System proven in operational environment
TRL 8	System complete and qualified
TRL 7	Integrated pilot system demonstrated
TRL 6	Prototype system verified
TRL 5	Laboratory testing of integrated system
TRL 4	Laboratory testing of prototype component or process
TRL 3	Critical function, proof of concept established
TRL 2	Technology concept and/or application formulated
TRL 1	Basic principles are observed and reported

CS-NLL PROJECT | STARTINGPOINT

Circular Flows – Next Level Logistics



1. Challenge to keep the Netherlands liveable and competitive in a more sustainable way.
2. Need for smarter planning and cooperation in transport is increasing:
 - sustainability → not more but smarter + smart handling of upcoming restrictions + circularity;
 - shortage of personnel.
3. Cooperation today is already pretty ok but can be improved, for example:
 - supply-chain cooperation with dominant party → efficient but possible advantage often with 1 or a few party(ies);
 - broker platforms → efficient exchange but usually concerns residual capacity and cargo surplus (trash can logistics);
 - Benelux distribution network organizations → already saves empty kilometers but usually static (zone) agreements.
4. More & more techniques are available that make smarter (joint) planning within reach for more companies.



CS-NLL PROJECT | THE ISSUE

Circular Flows – Next Level Logistics

How can we plan smarter (automatically) for both our own organization and for partners focused on sustainable - circular networks. Also: logistics as a fundamental utility for a sustainable - circular economy.

1. Opportunity spotting across planned schedules of several (transport)companies with the principle that everyone remains “the boss of their own body”. Also known as federated fleet (smartly collaboration of combined vehicle fleets).
2. Various existing techniques (AI - advanced planning algorithms - Digital Twins - ...) can be used for this → building on what is already there from involved knowledge partners and/or previous projects.
3. With extra attention for the planner, the planner must be happy about it - only then it will work. The happy planner!

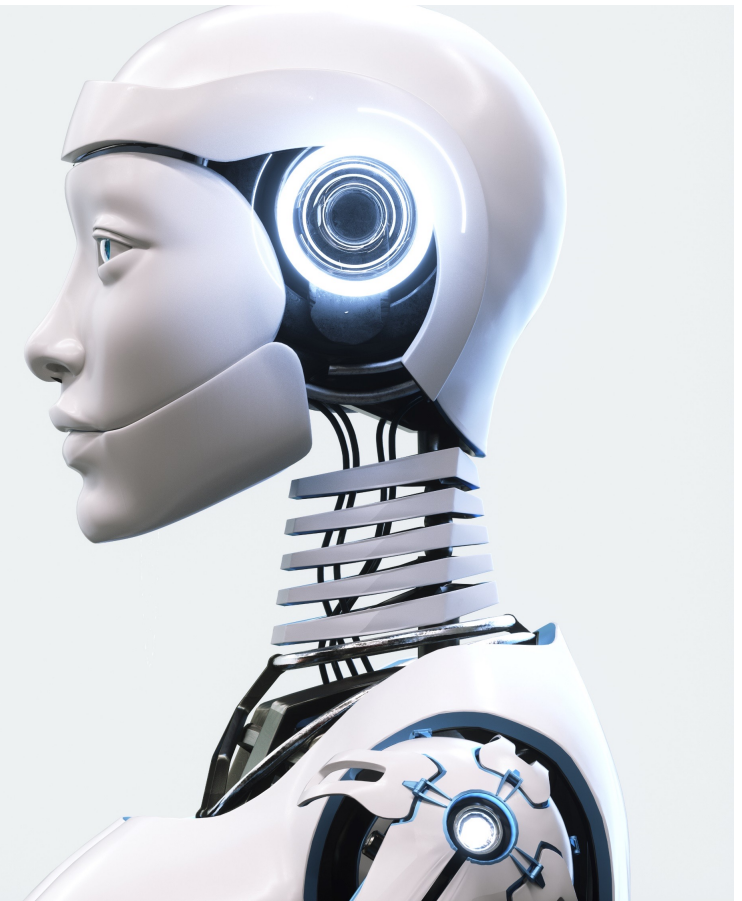
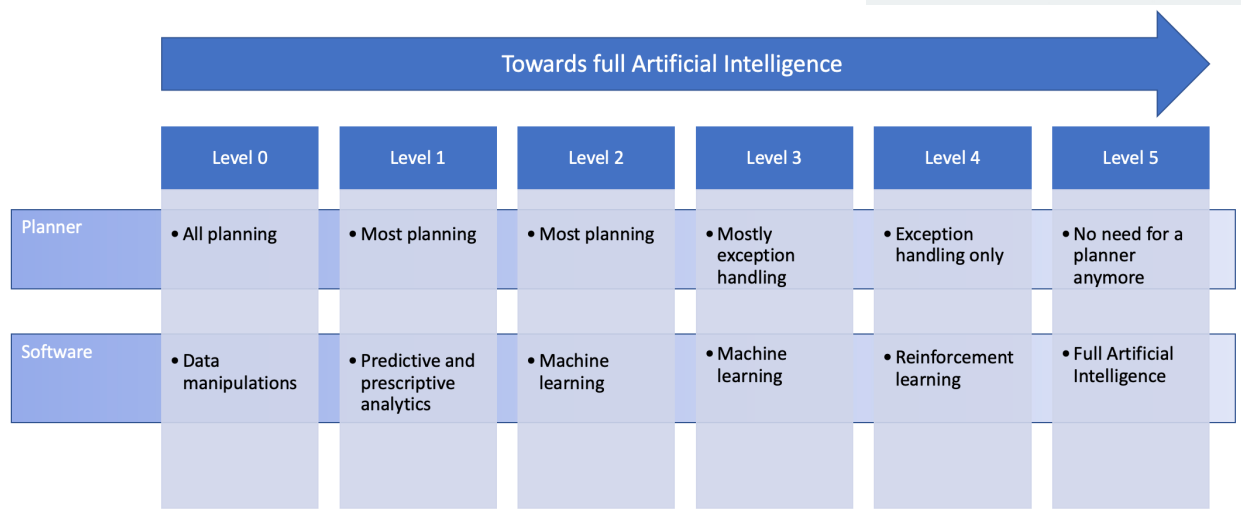


Future Proof Smart Logistics



Source: TNO

IS AI THE PLANNER OF THE FUTURE?



Source: TU/e – Tom van Woensel

CS-NLL PROJECT | THE PROJECT

Circular Flows – Next Level Logistics

- Project development by LCB and Midpoint Brabant
- Project coordination = LCB
- Knowledge partners: BUas, TNO, TU/e and Midpoint Brabant
- Number of participating companies: 12
- Duration of the project = 1-1-24 until 31-12-27.
- Subsidy providers: municipality of Tilburg, Province of North Brabant and Ministry of the Interior and Kingdom Relations

Project in 2 phases:

1. 4 cluster cases that (further) develop and test federated fleet POC tools, methods.
2. Online toolbox with which generic federated fleet POC tools and other input ((happy planner) workshop / training material / presentations / reports / ...) will be available



CS-NLL PROJECT | 4 CLUSTER-CASES

Circular Flows – Next Level Logistics

1. Multimodal: federated fleet truckplanning for inland terminals in Tilburg and Oosterhout.
2. Benelux distribution: Network Benelux cooperation (12 transport companies) to realize smart - more dynamic planning for freight exchange between network partners (via depot or directly).
3. Local for local (regional/city distribution): the local for local cooperation is aiming at smart bundling and combining flows in the delivery of regional fresh products in Central Brabant by own vehicles of involved farmers and/or centrally organized (outsourced) transport.
4. Circular networks: explore and, as far as possible, set up structures - methods - techniques - business models for circular operating centers and storage and transshipment stations for the benefit of circular supply chains, including identifying promising flows, creating jobs, developing skills, promoting social integration and increasing opportunities for entrepreneurship and innovation.



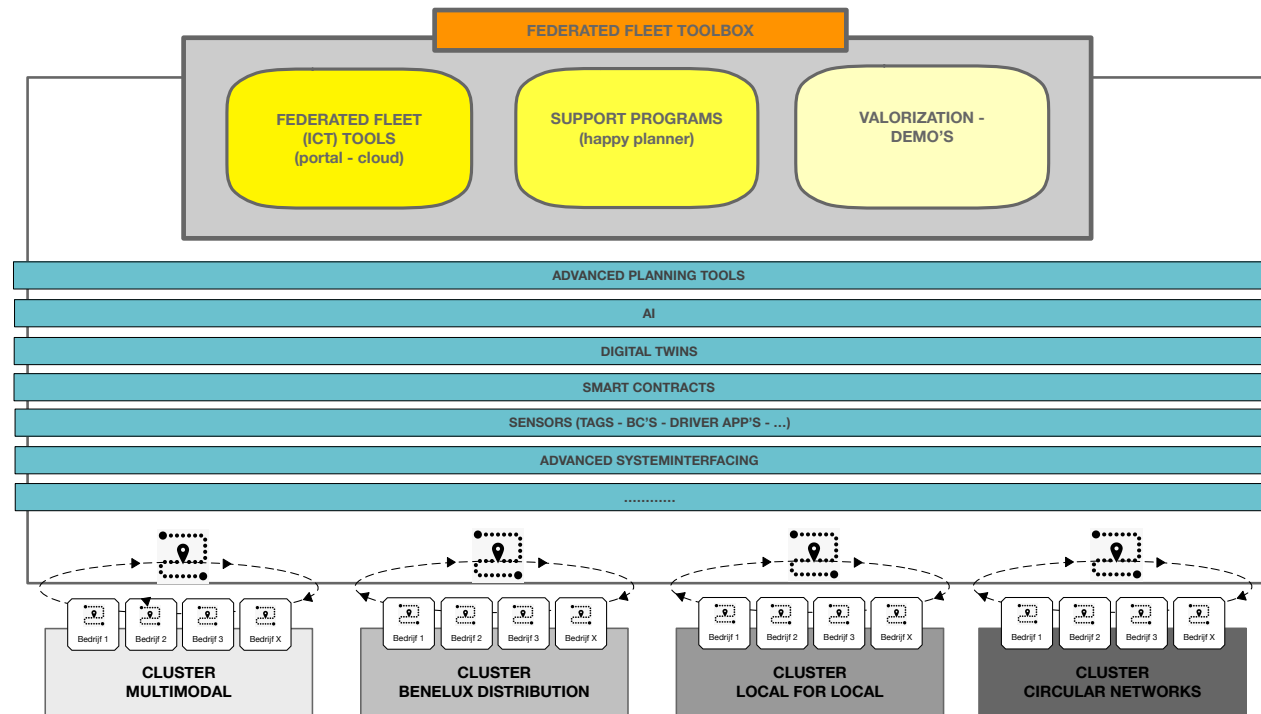
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CS-NLL PROJECT | EXAMPLE PREVIOUS EXECUTED PROJECT



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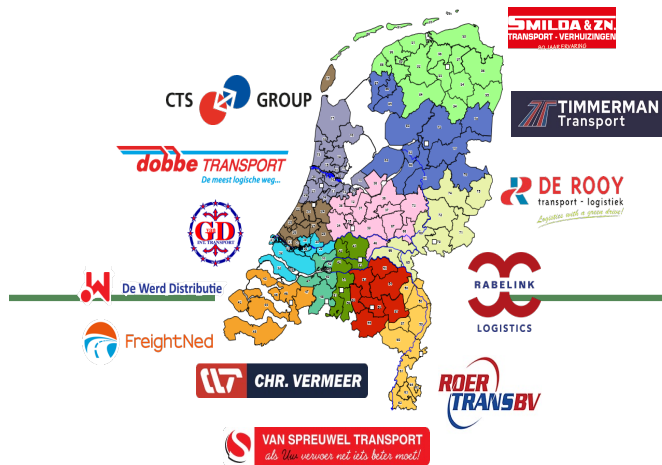
CS-NLL PROJECT | IN 1 VIEW



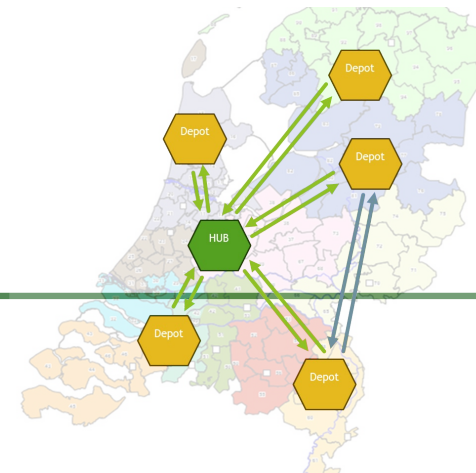
EXAMPLE OF DATA SCIENCE & AI STUDENT ASSIGNMENT (CS-NLL)

Objectives of a AI transportation planning system

1. Improve efficiency: minimize transportation times, costs, and fuel consumption.
2. Optimize resources: maximize the use of vehicles, personnel, and infrastructure.
3. Flexibility: adapt to changing circumstances such as traffic, weather conditions, and delivery requirements.
4. Customer satisfaction: deliveries on time and with high level service.
5. Sustainability: reduce environmental impact through more efficient route planning and fuel consumption.



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EXAMPLE OF DATA SCIENCE & AI STUDENT ASSIGNMENT (CS-NLL)

Components of the AI transportation planning system

1. Data Collection and Integration

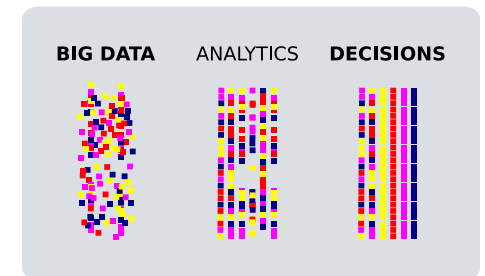
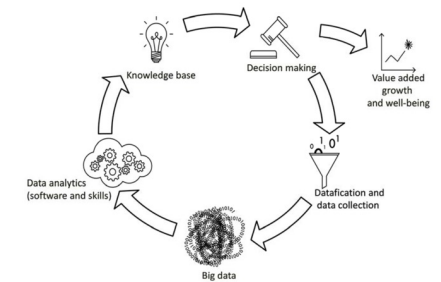
- Real-time data: traffic information, weather conditions, customer demand.
- Historical data: previous routes, delivery times.

2. Predictive Analysis

- Traffic forecasting: use machine learning to predict future traffic patterns.
- Demand forecasting: predict demand based on historical data and seasonal influences.

3. Optimization Algorithms

- Route optimization: use algorithms such as A* or genetic algorithms to plan the most efficient routes.
- Load planning: maximize vehicle loading to minimize costs.



EXAMPLE OF DATA SCIENCE & AI STUDENT ASSIGNMENT (CS-NLL)

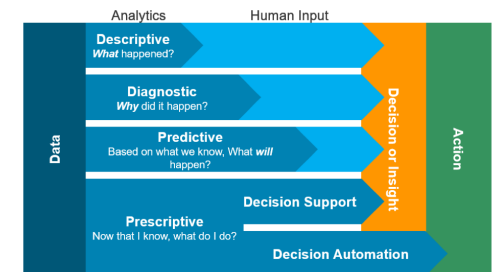
Components of the AI transportation planning system

4. Machine Learning and AI

- Adaptive planning: recognize learning patterns and adapt to changing conditions.
- Autonomous decision making: make autonomous decisions about route changes and priorities.

5. User Interface and Communication

- Real-time updates: provide drivers and planners with real-time updates about changes in routes or delivery times.
- Predictive alerts: warn of potential delays or issues based on predictive analysis.



Adapted from Gartner Report July 2015



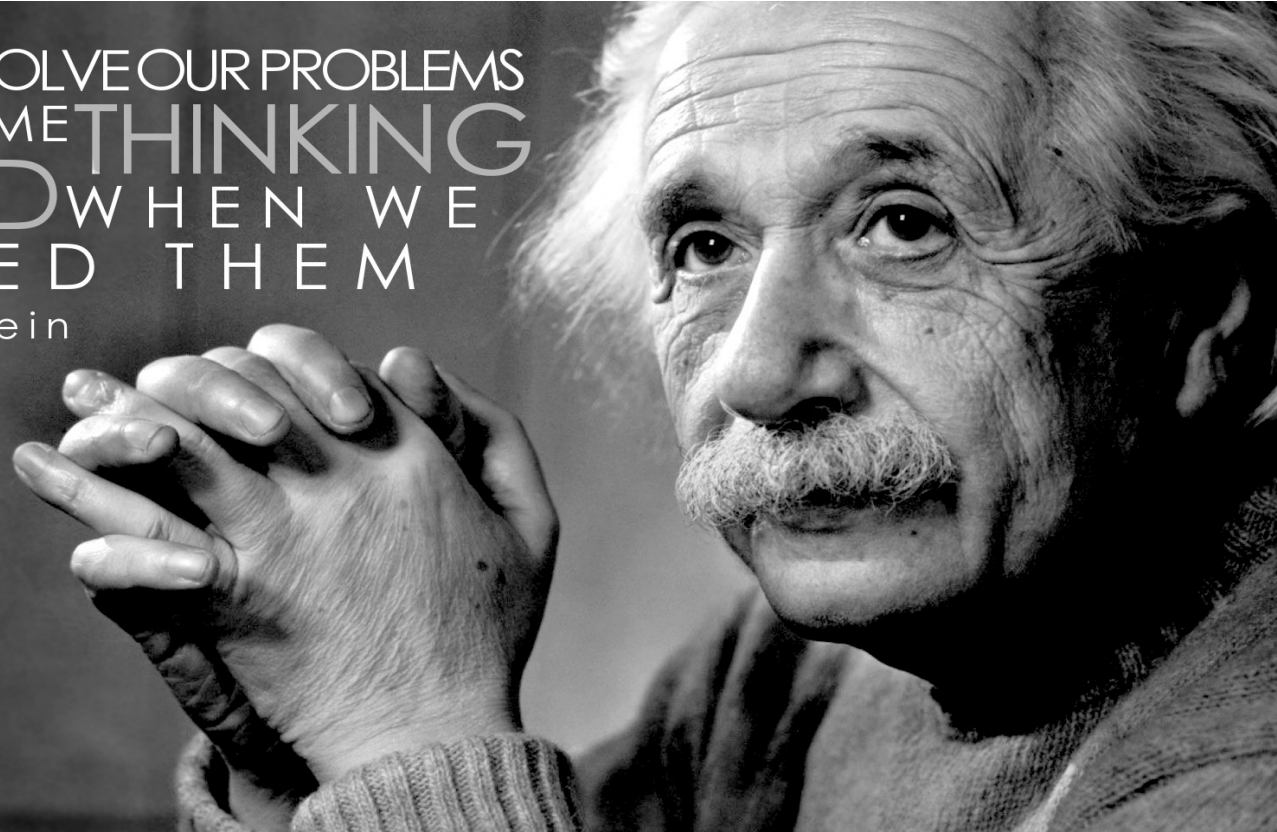
EXAMPLE OF DATA SCIENCE & AI STUDENT ASSIGNMENT (CS-NLL)

Suggested implementation steps of the AI transportation planning system

1. Proof of Concept: test the system in a limited environment to prove feasibility and effectiveness.
2. Gradual rollout: implement the system in stages, starting with less critical routes and expanding as confidence and experience grow.
3. Training and education: train staff and drivers to use the new system and provide them with the necessary skills to operate it.
4. Monitoring and evaluation: continue to monitor and evaluate the system to make improvements and identify new opportunities for optimization.
5. Iterative improvements: use user feedback and operational data to make continuous improvements to the system.

WE CANNOT SOLVE OUR PROBLEMS
WITH THE SAME THINKING
WE USED WHEN WE
CREATED THEM

-Albert Einstein



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